Optional Practice 3

- **Optional and ungraded:** No submission is required for Problems 1–2.
- **Point values:** Indicate suggested effort only and do not contribute to the course grade.
- **Instructor:** Bo Guo, University of Arizona
- **Semester:** Fall 2026

1. (15 points) Newton–Raphson method

Use the Newton–Raphson method to find an approximation of

$$\sqrt{10}$$

with an error

$$\epsilon < 10^{-8}.$$

You can choose your initial guess as 3. Show (without using the square root function of a calculator) that your answer is indeed within 10^{-8} of the true solution.

2. (55 points) One-dimensional transient advection–diffusion–reaction Equation

Consider the PDE:

$$\frac{\partial u}{\partial t} + V\frac{\partial u}{\partial x} - D\frac{\partial^2 u}{\partial x^2} + Ku = 0, \quad 0 < x < L,\ t > 0$$

Boundary and initial conditions:

$$u(0,t) = 1, \quad \frac{\partial u}{\partial x}(L,t) = 0, \quad u(x,0) = 0.$$

Tasks:

1. Write a finite difference approximation using:
   - variable upstream weighting (with weight α)
   - variably weighted Euler time-stepping (with weight θ)
2. Write a code to solve the resulting system of algebraic equations.
3. The number of grid points N_x should be a user input.

Use parameters:

$$L = 1, \quad V = 1, \quad D = 0.01, \quad K = 0.0001.$$

Investigate:

- Grid Peclet numbers:

$$Pe_G = \frac{V\,\Delta x}{D} \in \{0.1, 1, 2, 5, 10\}$$

- Different time steps Δt:
  - For conditionally stable cases, find the threshold Δt above which the numerical solution becomes unstable.
- Upstream weighting:

$$\alpha = 0,\ 0.5,\ 1$$

- Time weighting:

$$\theta = 0,\ 0.5,\ 1$$

Verification: Compare your numerical results with the analytical solution.

Hints

1. Use the semi-infinite analytical solution for comparison ($0 < x < \infty$). To ensure equivalence, choose a large enough domain $[0, L]$ or small simulation time t such that the Neumann boundary at $x = L$ is effectively zero flux.
2. Analytical solution (for semi-infinite domain):

$$u(x,t) = \frac{1}{2} e^{\frac{Vx}{2D}}$$

$$\left[e^{\frac{Vx}{2D} + x\sqrt{\frac{K}{D}}} \operatorname{erfc}\left(\frac{x}{2\sqrt{Dt}} + \sqrt{\frac{V^2 t}{4D} + Kt} \right) + e^{-\frac{Vx}{2D} - x\sqrt{\frac{K}{D}}} \operatorname{erfc}\left(\frac{x}{2\sqrt{Dt}} - \sqrt{\frac{V^2 t}{4D} + Kt} \right) \right]$$

where

$$\operatorname{erfc}(x) = \frac{2}{\sqrt{\pi}} \int_x^{+\infty} e^{-u^2}\, du$$

is the complementary error function (available in Python and MATLAB).

3. Final Project Proposal (Required Ungraded Milestone)

This proposal is a required milestone within the final project, even though Problems 1–2 are optional practice. Propose a term project problem that is:

- Challenging (several weeks of effort)
- A research problem in your field of interest
- Designed to fully leverage the relevant numerical and scientific machine-learning methods covered in the course

The final deliverables:

- Written report (5–10 pages) due Monday, December 14, 2026, at 5:30 pm
- Oral presentation tentatively scheduled for class Tuesday, December 1, 2026

Examples of topics:

- Contaminant transport with nonlinear reactions (e.g., biodegradation)

For the project proposal, provide:

- Your proposed topic
- A short description of the methods you intend to use and why they are appropriate
- Your plan for verifying, validating, or comparing the solution